

RAG System Evaluation: Hallucination & Retrieval Analysis

Case 1: Impossible Command Pipeline (Architectural Blindness)

Question: 'How can I use cp in a pipeline with tar and tee to create a bootable system backup?'

System Response: Suggested a command where 'cp' reads from a pipe: 'tar ... | tee ... | cp -a ...'.

Why it Hallucinated: The system treated 'cp' as a stream filter like 'tar -x'. It lacks a 'command ontology' to distinguish between tools that read stdin and tools that require file arguments. The LLM brute-forced an invention to satisfy the request for a single pipeline rather than admitting impossibility.

Case 2: False Causal Logic (Pattern-Matching Bias)

Question: 'Why does the cp command NOT preserve hard links by default, but the ln command does NOT create hard links to directories?'

System Response: Invented a causal link between symbolic link handling and hard link preservation that does not exist.

Why it Hallucinated: The LLM identified two concepts mentioned near each other in the source material and assumed they were related. It also performed 'semantic slot filling,' replacing technical terms about symbolic links with 'directories' to match the user's phrasing, fundamentally changing the technical meaning.

Case 3: Reference Fragmentation & Technical Omission (Combined)

Question: 'Explain the --indicator-style flag in ls.'

System Response: Omitted the 'file-type' option, claimed '-F' and '--indicator-style' should be used together (redundant), and cited a non-existent man-page section.

Why it Hallucinated:

1. Chunk Fragmentation: The technical documentation was split across chunk boundaries, causing the tail-end of the flag descriptions (like 'file-type') to be lost.
2. Overgeneralization: The LLM assumed two related options could be combined for additive effect.
3. Retrieval Pressure: The Bi-encoder retrieved the 'best guess' chunks, but they were incomplete, leading the LLM to fill gaps with fabricated citations.

Case 4: Internal Knowledge Leakage (Constraint Violation)

Questions: 'How do I install Windows 11...' or 'How to exclude .git in tar?'

System Response: Suggested valid commands (tar --exclude) that WERE NOT in the retrieved context.

The Failure: The LLM's internal training data (internal knowledge) overrode the 'Answer from context only' system constraint. It provided 'correct' Linux commands that were not citation-backed from the private store.

Architectural Mark: Fidelity failure. The system acts as a generic chatbot instead of a grounded assistant.